# Day 15: Model selection

## Warm up (quiz review)

`bikes_A` predicts `rides` from `temp_actual`, `windspeed`, and `weekend`. `bikes_B` adds `holiday`.

```
anova(bikes_A, bikes_B)
```

```
Analysis of Variance Table

Model 1: rides ~ temp_actual + windspeed + weekend
Model 2: rides ~ temp_actual + windspeed + weekend + holiday
  Res.Df       RSS Df Sum of Sq      F   Pr(>F)
1    496 755019445
2    495 743486506  1  11532939 7.6784 0.005799 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

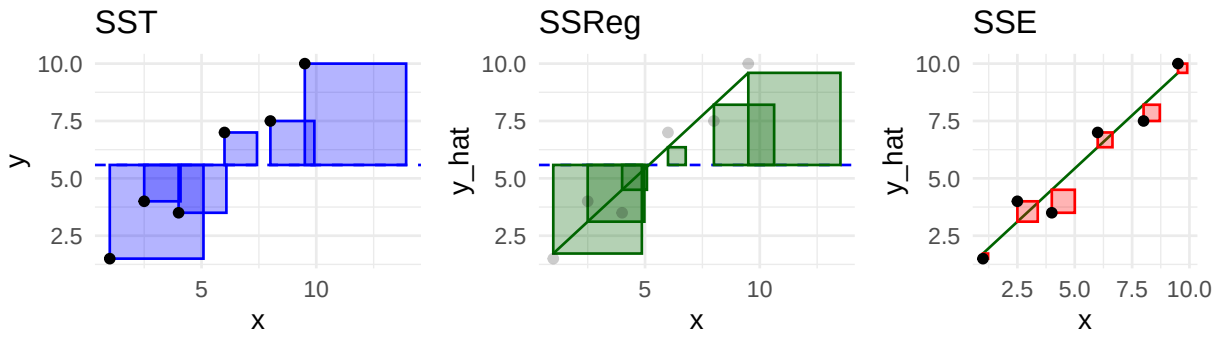
**1.** Which model is the full model? Write the null and alternative hypothesis for the test in the ANOVA table above.

**2.** What is $d$? What are $df_{full}$ and $SSE_{reduced}$?

**3.** State your conclusion in context.

**4.** What p-value would you expect for the `holidayyes` row of `summary(bikes_B)`? Why?

## Review: sums of squares




- Least squares makes the total area of the _____ squares as small as possible
- $R^2$: proportion of the _____ area (SST) covered by the _____ squares (SSReg)
- $R^2_{adj} = 1 - \frac{MSE}{MST}$: same idea, but each model is "charged" for the number of _____ it uses
- Overall $F$ test, $F = \frac{MSReg}{MSE}$: is the "average" _____ square bigger than the "average" _____ square?
- Extra sums of squares $F$ test: are the _____ squares for a larger model significantly smaller than for a smaller model?

## Today: building an explanatory model

**Question:** How are weekends and holidays associated with ridership, after accounting for the weather?

Two reasons to build a model:

- **Explanatory:** _____
- **Predictive:** _____

Today we're building an explanatory model. (Predictive models come after the quiz!)

**Which variables would you include to answer our question? Why?**

## A recipe for explanatory models

1. **Start with the question.** Which variables are you interested in? Which do you need to adjust for? These stay in the model, even if _____
2. **Look before you fit.** Check for predictors that _____
3. **Fit a *small* set of candidate models** (not every possible combination!)
4. **Compare the candidates**: _____ for nested models; _____ for any models

5. **Check the conditions** for the model you choose
6. **Interpret and justify** your choice

## Step 1: Our candidate models

- **Interested in:** weekend, holiday
- **Adjust for:** weather (temperature, windspeed, humidity)

| Model | Predictors |
|---|---|
| A | temp_actual + windspeed + weekend |
| B | A + holiday |
| C | B + humidity |
| D | C + temp_feel |
| E | temp_feel + humidity + weekend + holiday |

## Step 2: Look before you fit

```
bikes |>
  select(temp_actual, temp_feel, windspeed, humidity) |>
  cor() |>
  round(2)
```

```
            temp_actual temp_feel windspeed humidity
temp_actual        1.00      0.98     -0.11     0.27
temp_feel          0.98      1.00     -0.15     0.27
windspeed         -0.11     -0.15      1.00    -0.32
humidity           0.27      0.27     -0.32     1.00
```

Which pair of predictors could cause a problem? What problem?

## Step 3: Compare nested models with an $F$ test

**Which pairs are nested?** Circle one for each.

| Model 1 | Model 2 | Nested? |
|---|---|---|
| A | B | yes / no |
| B | D | yes / no |
| C | E | yes / no |
| A | C | yes / no |

$$F = \frac{(SSE_{reduced} - SSE_{full})/d}{SSE_{full}/df_{full}}$$

- Numerator: _____
- Denominator: _____
- A large $F$ means: _____

**Adding variables one at a time:**

| step | d | F | p-value |
|---|---|---|---|
| A to B: add holiday | 1 | 7.68 | 0.006 |
| B to C: add humidity | 1 | 85.43 | <0.001 |
| C to D: add temp_feel | 1 | 3.28 | 0.071 |

What do the $F$ tests say about each step?

## Careful: `temp_actual` and `temp_feel`

| model | term | estimate | std.error | p.value |
|---|---|---|---|---|
| C (without temp_feel) | temp_actual | 94.88 | 5.21 | <0.001 |
| D (with temp_feel) | temp_actual | 46.31 | 27.30 | 0.090 |
| D (with temp_feel) | temp_feel | 45.07 | 24.87 | 0.071 |

What happened to the standard error of `temp_actual`? Why? What should we do?

## When the $F$ test can't help

**Is** `bikes_E` **nested in** `bikes_C`**?** _____ Can we use `anova(bikes_E, bikes_C)`? _____

Two situations where the $F$ test can't help:

1. _____
2. _____

We need tools that work for *any* two models fit to the same data.

## Adjusted $R^2$

$$R^2_{adj} = 1 - \frac{MSE}{MST} = 1 - \frac{SSE/(n-(p+1))}{SST/(n-1)}$$

- $R^2$ always goes up when we add predictors; $R^2_{adj}$ _____
- can be used for nested **and** non-nested models
- no formal test or rule for "better enough"

| Model | Adj R2 |
|---|---|
| C: temp_actual + windspeed + weekend + holiday + humidity | 0.483 |
| E: temp_feel + humidity + weekend + holiday | 0.445 |

Which model does adjusted $R^2$ prefer? _____

## Information criteria: AIC and BIC

$$IC = \text{Measure of Fit} + \text{Penalty}$$

$$AIC = n\log(SSE/n) + 2(p+1) \qquad BIC = n\log(SSE/n) + \log(n)\cdot(p+1)$$

where $p$ = number of predictors. In R:

`AIC(model)` and `BIC(model)`.

- Smaller values are _____.
- When we add a predictor, the fit term gets _____ and the penalty gets _____.
- BIC penalizes more than AIC whenever $\log(n) >$ _____, i.e. $n \geq$ _____.

## Example: adjusted $R^2$ and $F$ tests by hand

A researcher fits four nested models to $n = 40$ observations. Fill in the table, then answer the questions.

| Model | Predictors | $p$ | SSE | $\hat{\sigma}^2 = SSE/(n-(p+1))$ | $R^2_{adj}$ |
|---|---|---|---|---|---|
| M0 | none | 0 | 980 | 25.13 | 0 |
| M1 | $x_1$ | 1 | 520 | 13.68 | 0.455 |
| M2 | $x_1, x_2$ | 2 | 410 | | |
| M3 | $x_1, x_2, x_3$ | 3 | 398 | | |

$R^2_{adj} = 1 - \hat{\sigma}^2/\hat{\sigma}^2_{M0}$ (since $MST = SSE_{M0}/(n-1)$).

**(a)** Which model is best by $R^2_{adj}$?

**(b)** Use an $F$ test to compare M3 to M2. State $d$ and $df_{full}$. (reminder: $F = \frac{(SSE_{reduced} - SSE_{full})/d}{SSE_{full}/df_{full}}$)

**(c)** Do the two tools agree? Which model would you choose?

## Step 4: Compare all of our candidates

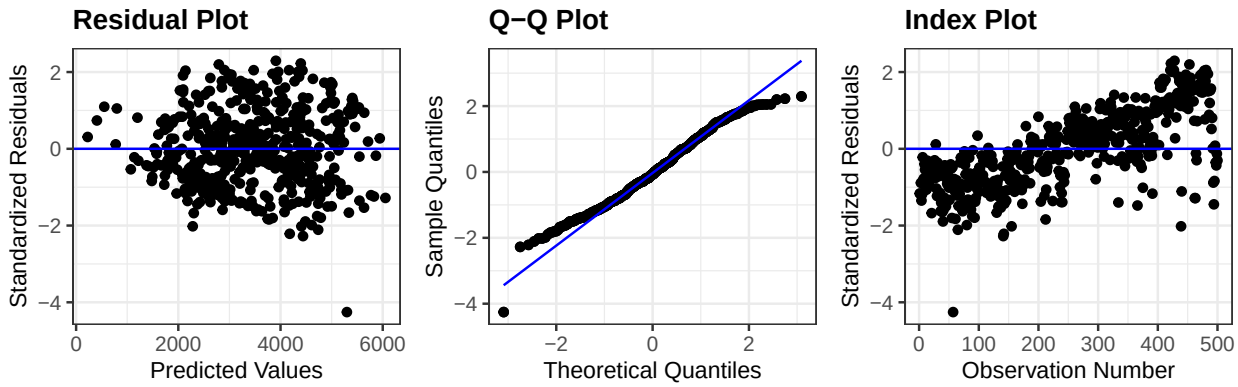
> **Activity**
>
> Work through Parts 1-3 of the R activity and record your results here.

| Model | $k$ | $R^2_{adj}$ | AIC | BIC |
|---|---|---|---|---|
| A: `temp_actual + windspeed + weekend` | | | | |
| B: A + `holiday` | | | | |
| C: B + `humidity` | | | | |
| D: C + `temp_feel` | | | | |
| E: `temp_feel + humidity + weekend + holiday` | | | | |

Which model would you choose to answer our question? Justify your choice.

> **Stop here**
>
> Let Amanda know when your group finishes!

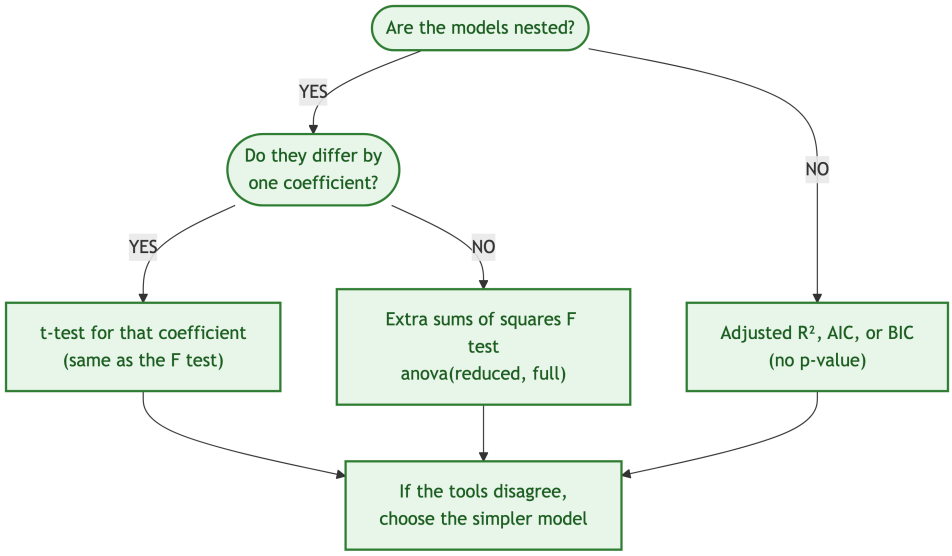
## Steps 5 and 6: Check and interpret




| term | estimate | conf.low | conf.high |
|---|---|---|---|
| weekendTRUE | -777 | -1000 | -554 |
| holidayyes | -1093 | -1751 | -435 |

**Example:** Using model C, answer our question in context.

## Comparing two models



## Summary

| Tool | Nested only? | Gives a p-value? | Notes |
|---|---|---|---|
| $F$ test | | | |
| Adjusted $R^2$ | | | |
| AIC | | | |
| BIC | | | |

- Use domain knowledge first. These tools compare candidate models; they don't replace thinking.
- **Justify your model choice!**